

Calculus of Single & Multiple Variables



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Overview

- Numbers, Infinity, Vectors and Functions
- Limit & Continuity
- Differentiation, Differentiability & Integration
- Numerical Differentiation & Integration
- Maxima & Minima
- Partial Differentiation & Gradient
- Hessian Matrix

Numbers and Infinity

- Natural Numbers
 - Closure under Addition
 - Concept of Identity under Addition: $a + x = a$ ($x = 0$)
 - No Additive Inverse
- Integers
 - Closure under Addition, Subtraction and Multiplication
 - No Inverse under Multiplication
- Rational Numbers $\left(\frac{p}{q}\right)$
 - Closed under Addition/Multiplication and Inverses/Identities Exist
 - Problem with Inverse of Exponentiation
- Concept of Irrational Numbers leading to Real Numbers
- Countable Infinity up to Rational Numbers
- Set of Irrational and Real Numbers have Uncountable Infinite Elements

Vectors and Functions

$$A = [a_1 \quad a_2 \quad \cdots \quad a_{n-1} \quad a_n]$$

Vector

Value Set

$$V = \{a_1, \dots a_i, \dots a_n\}$$

$$I = \{1, \dots i, \dots n\}$$

Index Set

Value Set (Range)

$$y = f(x) \Rightarrow f: \mathbb{R}^1 \rightarrow \mathbb{R}^1$$

Index Set (Domain)

Limits

The **Limit** of a function $f(x)$ approaches some value L as the input x approaches some value c .

$$\lim_{x \rightarrow c} f(x) = L$$

Limit of f of x equals L as x approaches c

Limits are essential to calculus and are used to define continuity, derivatives, and integrals.

Continuity

A continuous function does not have any abrupt changes in value

$$\lim_{x \rightarrow c-h} f(x) = f(c) = \lim_{x \rightarrow c+h} f(x)$$

h is infinitesimally small number

Differentiation

The derivative of a function of a real variable measures the sensitivity to change of the function value (output value) with respect to a change in its argument (input value).

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

A differentiable function of one real variable is a function whose derivative exists at each point in its domain.

Differentiation

$$f(x) = c$$

$$f'(x) = ?$$

$$f(x) = x$$

$$f'(x) = ?$$

$$f(x) = ax$$

$$f'(x) = ?$$

$$f(x) = x^n$$

$$f'(x) = ?$$

Differentiation

$$f(x) = c$$

$$f'(x) = 0$$

$$f(x) = x$$

$$f'(x) = 1$$

$$f(x) = ax$$

$$f'(x) = a$$

$$f(x) = x^n$$

$$f'(x) = nx^{n-1}$$

Differentiation

$$f(x) = g(x)h(x)$$

$$f'(x) = ?$$

$$f(x) = \frac{g(x)}{h(x)}$$

$$f'(x) = ?$$

Differentiation

$$f(x) = g(x)h(x)$$

$$f'(x) = g'(x)h(x) + g(x)h'(x)$$

$$f(x) = \frac{g(x)}{h(x)}$$

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{\{h(x)\}^2}$$

Differentiation

$$f(x) = e^x$$

$$f'(x) = ?$$

$$f(x) = \log(x)$$

$$f'(x) = ?$$

$$f(x) = \sin(x)$$

$$f'(x) = ?$$

$$f(x) = \tan^{-1}(x)$$

$$f'(x) = ?$$

Differentiation

$$f(x) = e^x$$

$$f'(x) = e^x$$

$$f(x) = \log(x)$$

$$f'(x) = \frac{1}{x}$$

$$f(x) = \cos(x)$$

$$f'(x) = -\sin(x)$$

$$f(x) = \tan^{-1}(x)$$

$$f'(x) = \frac{1}{1+x^2}$$

Differentiation

$$f(x) = \phi(\psi(\zeta(x)))$$

$$f'(x) = \phi'(\psi(\zeta(x))) \psi'(\zeta(x)) \zeta'(x)$$

Differentiation

$$f(x) = \{g(x)\}^n$$

$$f'(x) = ?$$

$$f(x) = a^{\{g(x)\}}$$

$$f'(x) = ?$$

$$f(x) = \sqrt{g(x)}$$

$$f'(x) = ?$$

$$f(x) = \sin\{g(x)\}$$

$$f'(x) = ?$$

Differentiation

$$f(x) = \{g(x)\}^n$$

$$f'(x) = n g'(x) \{g(x)\}^{n-1}$$

$$f(x) = a^{\{g(x)\}}$$

$$f'(x) = a^{\{g(x)\}} \log(a) g'(x)$$

$$f(x) = \sqrt{g(x)}$$

$$f'(x) = \frac{g'(x)}{2\sqrt{g(x)}}$$

$$f(x) = \sin\{g(x)\}$$

$$f'(x) = g'(x) \cos\{g(x)\}$$

Differentiation

$$\textit{Sigmoid}(x) = f(x) = \frac{1}{1 + e^{-x}}$$

$$f'(x) = ?$$

Differentiation

$$f(x) = \frac{1}{1 + e^{-x}} = (1 + e^{-x})^{-1}$$

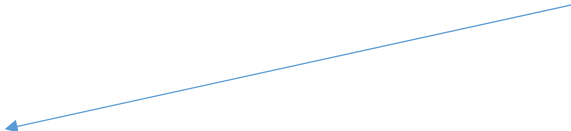
$$f'(x) = (-1) \times (1 + e^{-x})^{-2} \times (e^{-x}) \times (-1)$$

$$f'(x) = \frac{e^{-x}}{(1 + e^{-x})^2} = \frac{1 + e^{-x} - 1}{(1 + e^{-x})^2}$$

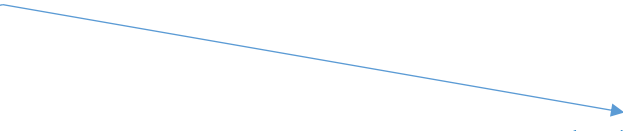
$$f'(x) = \frac{1}{1 + e^{-x}} - \frac{1}{(1 + e^{-x})^2} = f(x) - \{f(x)\}^2$$

Numerical Differentiation

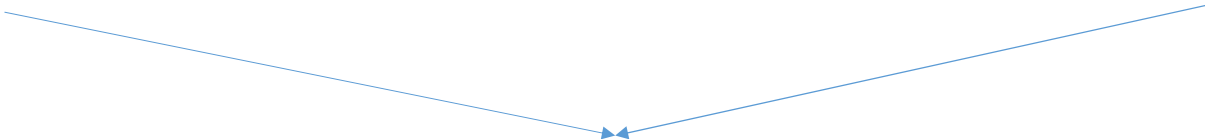
$$\frac{df(x)}{dx} = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$


$$\widehat{f'(a)} = \frac{f(a+h) - f(a)}{h}$$

Forward Differentiation


$$\widehat{f'(a)} = \frac{f(a) - f(a-h)}{h}$$

Backward Differentiation


$$\widehat{f'(a)} = \frac{f(a+h) - f(a-h)}{2h}$$

Numerical Differentiation

$$\widehat{f'(a)} = \frac{f(a+h) - f(a-h)}{2h}$$

$$f(x) = 2x^2 + 3x + 5$$

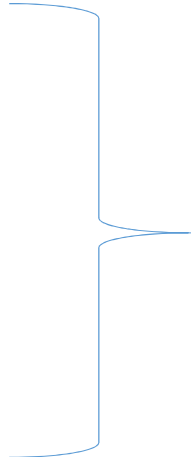
$$f'(3) = ? \quad h = 0.001$$

Numerical Differentiation

$$f(x) = 2x^2 + 3x + 5 \quad \Rightarrow \quad f'(x) = 4x + 3 \quad \Rightarrow \quad f'(3) = 15$$

$$f'(3) = \frac{f(3 + 0.001) - f(3 - 0.001)}{2 \times 0.001}$$

$$f'(3) = \frac{32.015002 - 31.985002}{0.002} = 15$$


$$h = 0.001$$

$$\widehat{f'(a)} = \frac{f(a + h) - f(a - h)}{2h}$$

Numerical Differentiation

$$f(x) = 5 + x \sin(3x + 2)$$

$$f'(x) = ?$$

$$f'(1) = ?$$

$$h = 0.001$$

Direct Differentiation

$$f(x) = 5 + x\sin(3x + 2)$$

$$f'(x) = \sin(3x + 2) + 3x\cos(3x + 2)$$

$$f'(1) = \sin(3 \times 1 + 2) + 3 \times 1 \times \cos(3 \times 1 + 2)$$

$$f'(1) = \sin(5) + 3\cos(5) = -0.1079$$

Numerical Differentiation

$$f(x) = 5 + x \sin(3x + 2)$$

$$f(1 + h) = 5 + (1.001) \times \sin(3 \times (1.001) + 2)$$

$$f(1.001) = 5 + 1.001 \sin(5.001) = 4.040973$$

$$f(1 - h) = 5 + (0.999) \times \sin(3 \times (0.999) + 2)$$

$$f(0.999) = 5 + 0.999 \sin(4.997) = 4.041189$$

$$h = 0.001$$

$$\widehat{f'(1)} = \frac{f(1 + h) - f(1 - h)}{2h} = \frac{4.040973 - 4.041189}{2 \times 0.001} = -0.108$$

Integration

Consider the function $f(x)$ of $x \in \mathbb{R}^1$. The Definite Integral over an interval $[a, b] \subset \mathbb{R}^1$

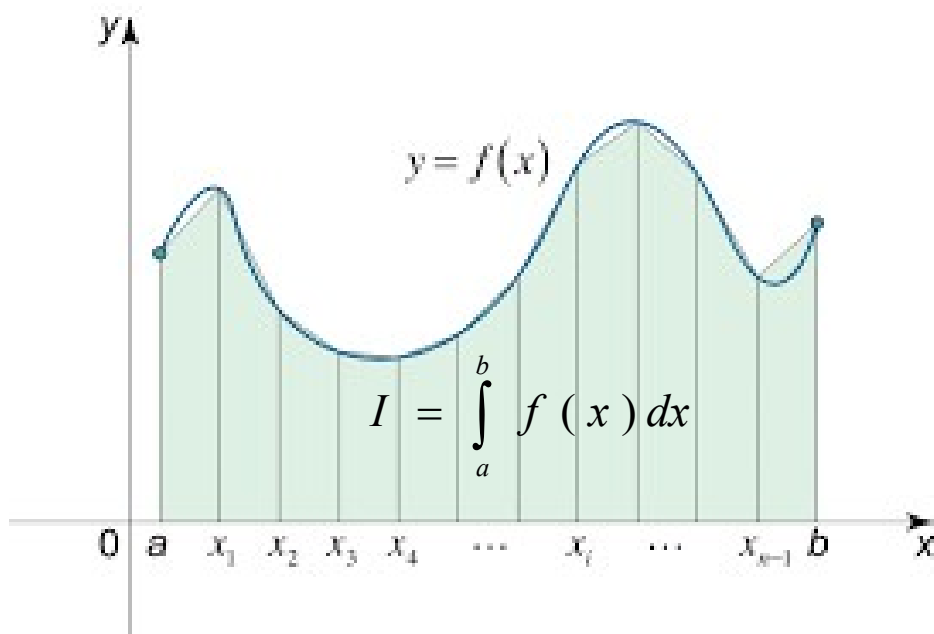
$$\int_a^b f(x) dx$$

This is interpreted as the signed area in the xy -plane bounded by $f(x)$, the x -axis and the vertical lines $x = a$ and $x = b$

Numerical Integration

Trapezoidal Rule: Interval $[a, b]$ is uniformly divided into n parts.

Approximate Value of Integral as Sum of Areas of Trapeziums formed in each part



$$h = \frac{b - a}{n}$$

$$A_i = \frac{h}{2} (y_i + y_{i+1})$$

$$I = \frac{h}{2} \sum_{i=0}^{n-1} (y_i + y_{i+1})$$

Taylor Series & Maclaurin's Theorem

The function $f(x)$ is infinitely differentiable at $x = a$

$$f(a + h) = \sum_{k=0}^{\infty} \frac{h^k f^{(k)}(a)}{k!} = f(a) + hf^{(1)}(a) + \frac{h^2 f^{(2)}(a)}{2!} + \dots$$

Substitutions: h by x and a by 0

$$f(x) = \sum_{k=0}^{\infty} \frac{x^k f^{(k)}(0)}{k!} = f(0) + xf^{(1)}(0) + \frac{x^2 f^{(2)}(0)}{2!} + \dots$$

$$f^{(1)}(x) \equiv f'(x); f^{(2)}(x) \equiv f''(x)$$

Polynomial Expansion of Functions

$$f(x) = \sin(x) - \cos(x)$$



$$f(x) = a_0 + a_1x + a_2x^2 + \cdots \infty$$

Polynomial Expansion of Functions

$$f(x) = \sin(x) - \cos(x) \quad \Rightarrow \quad f(0) = -1$$

$$f^{(1)}(x) = \cos(x) + \sin(x) \quad \Rightarrow \quad f^{(1)}(0) = 1$$

$$f^{(2)}(x) = -\sin(x) + \cos(x) \quad \Rightarrow \quad f^{(2)}(0) = 1$$

$$f^{(3)}(x) = -\cos(x) - \sin(x) \quad \Rightarrow \quad f^{(3)}(0) = -1$$

$$f^{(4)}(x) = \sin(x) - \cos(x) \quad \Rightarrow \quad f^{(4)}(0) = -1$$

Polynomial Expansion of Functions

$$f(x) = \sum_{k=0}^{\infty} \frac{x^k f^{(k)}(0)}{k!} = f(0) + x f^{(1)}(0) + \frac{x^2 f^{(2)}(0)}{2!} + \dots$$

$$f(x) = \sin(x) - \cos(x)$$

$$f(x) = -1 + x + \frac{x^2}{2!} - \frac{x^3}{3!} - \frac{x^4}{4!} + \dots \infty$$

Tracking with Motion Model

$$s = s_0 + u\Delta t + \frac{1}{2}f\Delta t^2$$

Equation of Motion

$$x(t + \Delta t) = x(t) + (\Delta t)x'(t) + \frac{1}{2}(\Delta t)^2x''(t)$$

Taylor's Series

Tracking with Motion Model

Any Physical
Quantity

δt : Time Domain Sampling Interval

$$\frac{dp}{dt} \approx \frac{p(n \times \delta t) - p((n - 1) \times \delta t)}{\delta t} = \frac{p_n - p_{n-1}}{\delta t}$$



Position

Velocity $\longrightarrow$

$$x'(t) \approx \frac{x_n - x_{n-1}}{\delta t} = v_n$$

Tracking with Motion Model

δt : Time Domain Sampling Interval

$$\frac{dp}{dt} \approx \frac{p_n - p_{n-1}}{\delta t} \quad \longrightarrow \quad x'(t) \approx \frac{x_n - x_{n-1}}{\delta t} = v_n$$

$$x''(t) \approx ?$$

Acceleration

Tracking with Motion Model

δt : Time Domain Sampling Interval

$$x''(t) \approx \frac{v_n - v_{n-1}}{\delta t} = a_n$$

$$a_n = \frac{\frac{x_n - x_{n-1}}{\delta t} - \frac{x_{n-1} - x_{n-2}}{\delta t}}{\delta t} = \frac{x_n - 2x_{n-1} + x_{n-2}}{(\delta t)^2}$$

Tracking with Motion Model

$$x(t + \Delta t) = x(t) + (\Delta t)x'(t) + \frac{1}{2}(\Delta t)^2x''(t)$$

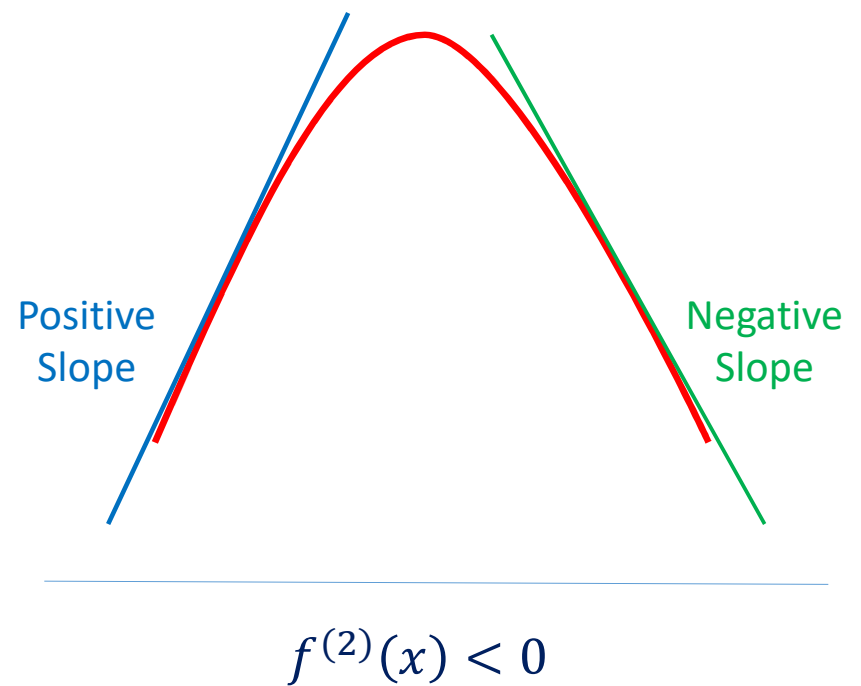
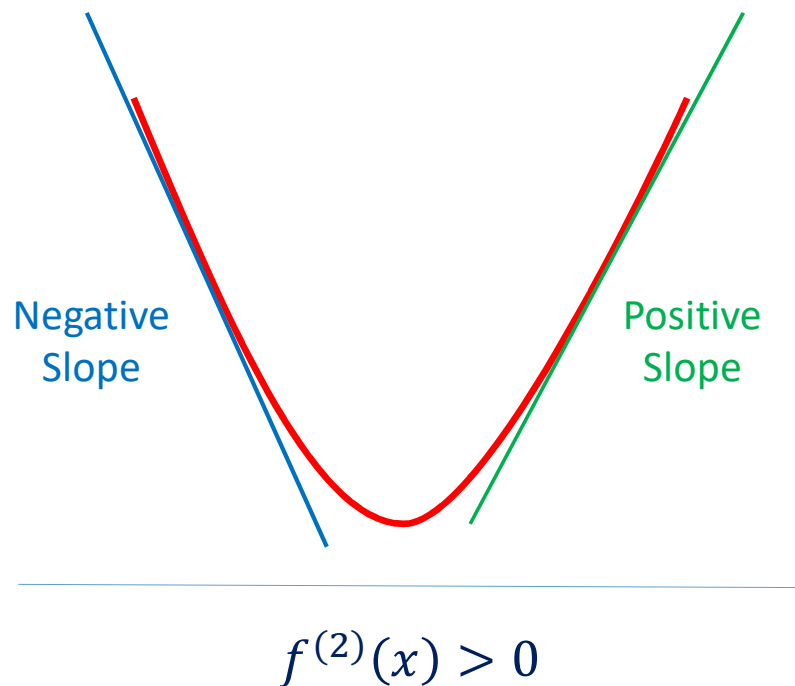


$$\hat{x}_n = x_{n-1} + (\delta t) \frac{x_{n-1} - x_{n-2}}{\delta t} + \frac{1}{2}(\delta t)^2 \frac{x_{n-1} - 2x_{n-2} + x_{n-3}}{(\delta t)^2}$$



$$\hat{x}_n = 2.5x_{n-1} - 2x_{n-2} + 0.5x_{n-3}$$

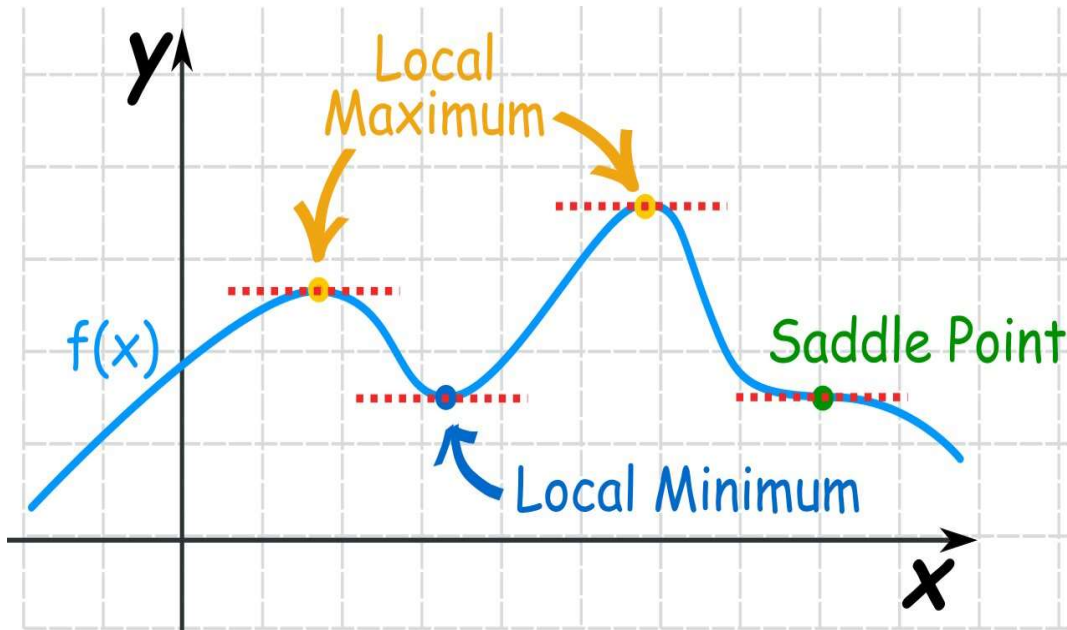
Role of $f^{(2)}(x)$ in Maxima-Minima: $y = f(x)$



Maxima & Minima

Differentiable function $f(x)$ has **local maxima** at $x = a$ if $f'(a) = 0$ and $f''(a) < 0$

Differentiable function $f(x)$ has **local minima** at $x = a$ if $f'(a) = 0$ and $f''(a) > 0$



Differentiable function $f(x)$ has a **saddle point** at $x = a$ if $f'(a) = 0$ and $f''(a) = 0$

Function of Multiple Variables

Consider Function $y = f(\mathbf{x}); \mathbf{x} \in \mathbb{R}^n, y \in \mathbb{R}^1$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$


$$y = f(x_1, \dots, x_i, \dots, x_n)$$

Partial Derivatives

A partial derivative of a function of several variables is its derivative with respect to one of those variables, with the others held constant.

$$y = f(x_1, \dots, x_i, \dots, x_n)$$

$$\frac{\partial f}{\partial x_i} = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_i, \dots, x_n)}{h}$$

Partial Differentiation

$$y = f(x_1, x_2) = x_1^2 + 2x_1 + 3x_1x_2 + 2x_2 + x_2^2$$

$$\frac{\partial y}{\partial x_1} = ?$$

$$\frac{\partial y}{\partial x_2} = ?$$

Partial Differentiation

$$y = f(x_1, x_2) = x_1^2 + 2x_1 + 3x_1x_2 + 2x_2 + x_2^2$$

$$\frac{\partial y}{\partial x_1} = 2x_1 + 2 + 3x_2$$

$$\frac{\partial y}{\partial x_2} = 3x_1 + 2 + 2x_2$$

Partial Differentiation

$$y = f(x_1, x_2, x_3) = x_1 \sin(x_2 x_3) + x_2 e^{x_1 + x_3} + \sqrt{x_2 \log(x_3)}$$

$$\frac{\partial y}{\partial x_1} = ?$$

$$\frac{\partial y}{\partial x_2} = ?$$

$$\frac{\partial y}{\partial x_3} = ?$$

Partial Differentiation

$$y = f(x_1, x_2, x_3) = x_1 \sin(x_2 x_3) + x_2 e^{x_1 + x_3} + \sqrt{x_2 \log(x_3)}$$

$$\frac{\partial y}{\partial x_1} = \sin(x_2 x_3) + x_2 e^{x_1} e^{x_3} = \sin(x_2 x_3) + x_2 e^{x_1 + x_3}$$

$$\frac{\partial y}{\partial x_2} = x_1 x_3 \cos(x_2 x_3) + e^{x_1 + x_3} + \frac{1}{2} \sqrt{\frac{\log(x_3)}{x_2}}$$

$$\frac{\partial y}{\partial x_3} = x_1 x_2 \cos(x_2 x_3) + x_2 e^{x_1 + x_3} + \frac{1}{2x_3} \sqrt{\frac{x_2}{\log(x_3)}}$$

Numerical Differentiation

$$\frac{\partial f}{\partial x_i} = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_i, \dots, x_n)}{h}$$



$$\frac{\widehat{\partial f}}{\partial x_i} \approx \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_i, \dots, x_n)}{h}$$

Numerical Differentiation

$$y = f(x_1, x_2) = x_1^2 + 2x_1 + 3x_1x_2 + 2x_2 + x_2^2$$

$$f'(1, -1) = ?$$

Numerical Partial Differentiation

$$f(x_1, x_2) = x_1^2 + 2x_1 + 3x_1x_2 + 2x_2 + x_2^2$$

$$f(1, -1) = -1$$

$$f(1 + 0.001, -1) = -0.998999$$

$$f(1, -1 + 0.001) = -0.996999$$

$$\frac{\partial y}{\partial x_1}(x_1 = 1) = \frac{-0.998999 - (-1)}{0.001} = 1.001$$

$$\frac{\partial y}{\partial x_2}(x_2 = -1) = \frac{-0.996999 - (-1)}{0.001} = 3.001$$

Gradient of a Function

Consider Function $y = f(\mathbf{x}); \mathbf{x} \in \mathbb{R}^n, y \in \mathbb{R}^1$

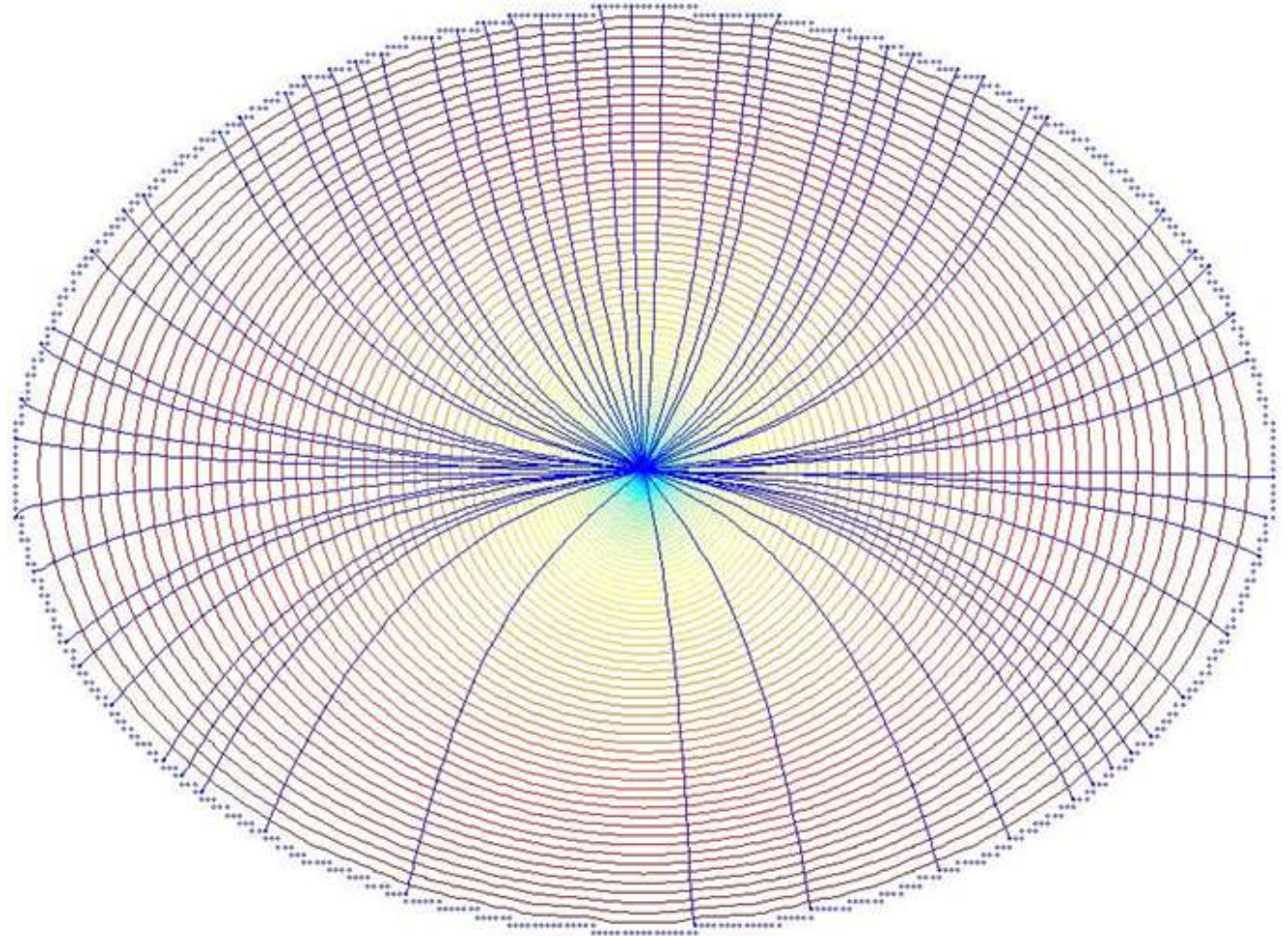
$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Grad of $f(\mathbf{x})$ w.r.t. $\mathbf{x}$

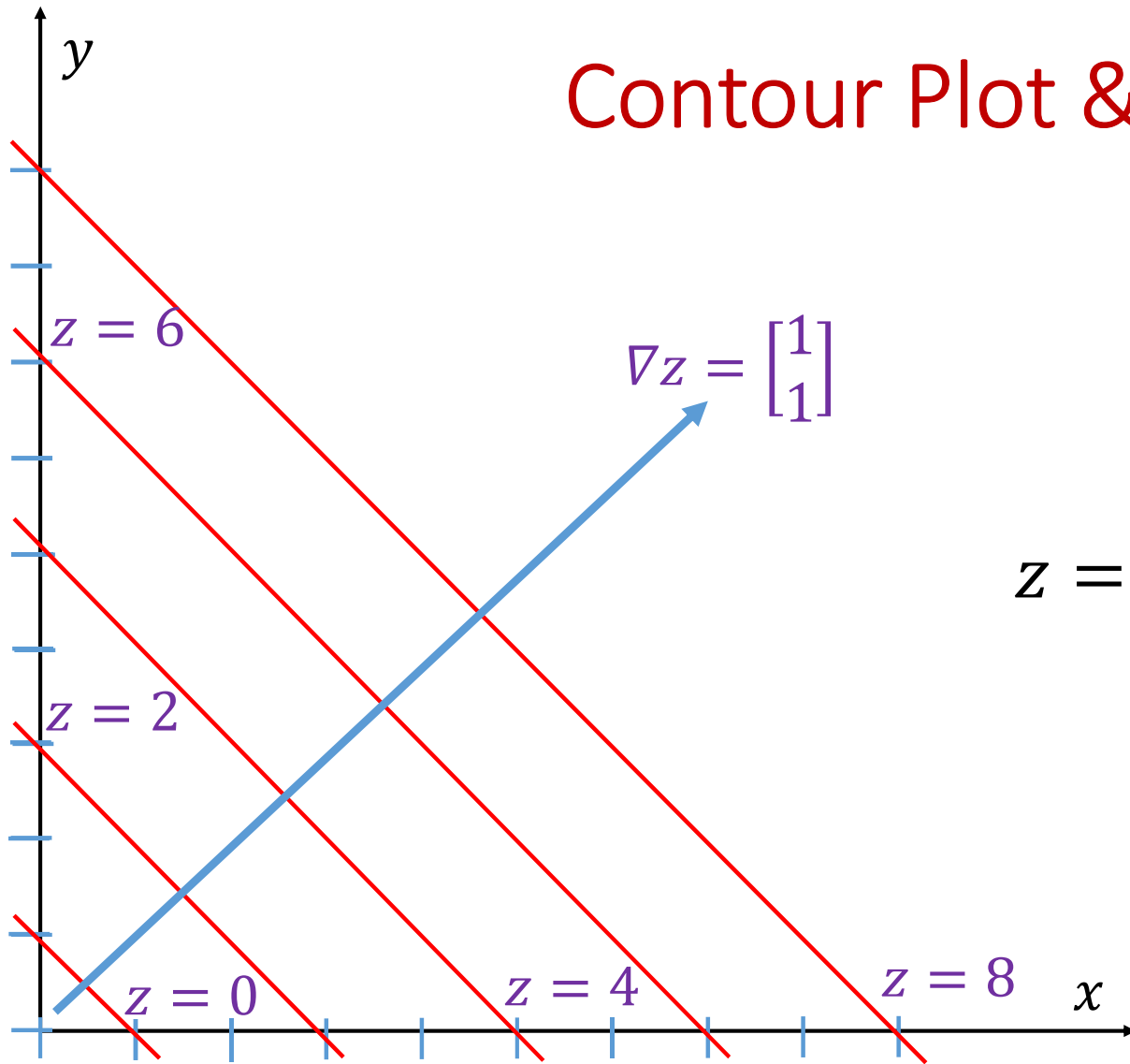
$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

Gradient and Contours

$$\nabla_x f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$



Contour Plot & Gradient



$$z = f(x, y) = x + y - 1$$

Gradient: A Few Basic Relations

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad y = \mathbf{b}^T \mathbf{x}$$

$$\nabla_{\mathbf{x}} y = ?$$

Gradient: A Few Basic Relations

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad y = \mathbf{b}^T \mathbf{x}$$

$$\nabla_{\mathbf{x}} y = \mathbf{b}$$

Gradient: A Few Basic Relations

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad y = \mathbf{x}^T \mathbf{b}$$

$$\nabla_{\mathbf{x}} y = ?$$

Gradient: A Few Basic Relations

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad y = \mathbf{x}^T \mathbf{b}$$

$$\nabla_{\mathbf{x}} y = \mathbf{b}$$

Gradient: A Few Basic Relations

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

$$y = \mathbf{x}^T A \mathbf{x}$$

$$\nabla_{\mathbf{x}} y = ?$$

Try with $n=2$

Illustration with n=2

$$y = \mathbf{x}^T A \mathbf{x}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$



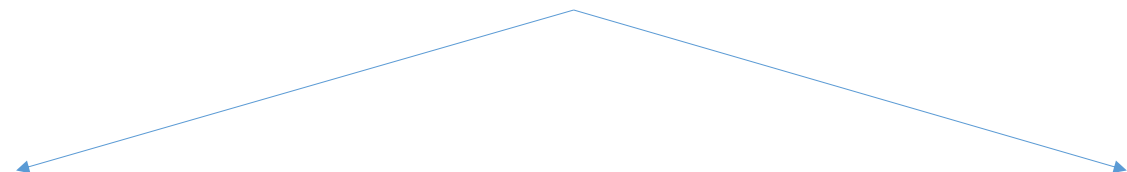
$$y = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} ax_1 + bx_2 \\ cx_1 + dx_2 \end{bmatrix}$$



$$y = ax_1^2 + (b + c)x_1x_2 + dx_2^2$$

$$\nabla_{\mathbf{x}} y = ?$$

Illustration with $n=2$

$$y = ax_1^2 + (b + c)x_1x_2 + dx_2^2$$


$$\frac{\partial y}{\partial x_1} = 2ax_1 + (b + c)x_2$$

$$\frac{\partial y}{\partial x_2} = (b + c)x_1 + 2dx_2$$

$$\nabla_x y = \begin{bmatrix} \frac{\partial y}{\partial x_1} \\ \frac{\partial y}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2ax_1 + (b + c)x_2 \\ (b + c)x_1 + 2dx_2 \end{bmatrix}$$

Illustration with $n=2$

$$\nabla_x y = \begin{bmatrix} 2ax_1 + (b+c)x_2 \\ (b+c)x_1 + 2dx_2 \end{bmatrix}$$



$$\nabla_x y = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a & c \\ b & d \end{bmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



$$\nabla_x y = (A + A^T)x$$

Gradient: A Few Basic Relations

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$$

$$y = \mathbf{x}^T A \mathbf{x}$$

$$\nabla_{\mathbf{x}} y = (A + A^T) \mathbf{x}$$

The Hessian Matrix

$$y = f(\mathbf{x}); \mathbf{x} \in \mathbb{R}^n, y \in \mathbb{R}^1$$

$$\nabla_{\mathbf{x}}^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

The Hessian Matrix

$$y = f(x_1, x_2)$$

$$\nabla_x^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 x_2} \\ \frac{\partial^2 f}{\partial x_2 x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_2} \right) \\ \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_1} \right) & \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_2} \right) \end{bmatrix}$$

Hessian Matrix

$$y = f(x_1, x_2) = x_1^2 + 2x_1 + 3x_1x_2 + 2x_2 + x_2^2$$

$$\nabla_x^2 f = ?$$

Hessian Matrix

$$y = f(x_1, x_2) = x_1^2 + 2x_1 + 3x_1x_2 + 2x_2 + x_2^2$$

$$\frac{\partial f}{\partial x_1} = 2x_1 + 2 + 3x_2$$

$$\frac{\partial^2 f}{\partial x_2 \partial x_1} = ?$$

$$\frac{\partial^2 f}{\partial x_1^2} = ?$$

Hessian Matrix

$$y = f(x_1, x_2) = x_1^2 + 2x_1 + 3x_1x_2 + 2x_2 + x_2^2$$

$$\frac{\partial f}{\partial x_1} = 2x_1 + 2 + 3x_2$$

$$\frac{\partial^2 f}{\partial x_1^2} = 2$$

$$\frac{\partial^2 f}{\partial x_2 \partial x_1} = \frac{\partial}{\partial x_2} \left(\frac{\partial f}{\partial x_1} \right) = 3$$

Hessian Matrix

$$y = f(x_1, x_2) = x_1^2 + 2x_1 + 3x_1x_2 + 2x_2 + x_2^2$$

$$\frac{\partial f}{\partial x_2} = 3x_1 + 2 + 2x_2$$

$$\frac{\partial^2 f}{\partial x_1 x_2} = ?$$

$$\frac{\partial^2 f}{\partial x_2^2} = ?$$

Hessian Matrix

$$y = f(x_1, x_2) = x_1^2 + 2x_1 + 3x_1x_2 + 2x_2 + x_2^2$$

$$\frac{\partial f}{\partial x_2} = 3x_1 + 2 + 2x_2$$

$$\frac{\partial^2 f}{\partial x_2^2} = 2$$

$$\frac{\partial^2 f}{\partial x_1 x_2} = \frac{\partial}{\partial x_1} \left(\frac{\partial f}{\partial x_2} \right) = 3$$

Hessian Matrix

$$y = f(x_1, x_2) = x_1^2 + 2x_1 + 3x_1x_2 + 2x_2 + x_2^2$$

$$\nabla_x^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 x_2} \\ \frac{\partial^2 f}{\partial x_2 x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$$

Numerical Problem

Consider the following function

$$f(\mathbf{x}) = x_1 - x_2 + 2x_1x_2 + 2x_1^2 + x_2^2$$

Evaluate: $\nabla f(\mathbf{a}), \nabla^2 f(\mathbf{a})$

Where $\mathbf{a} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$

Numerical Problem

$$f(\mathbf{x}) = x_1 - x_2 + 2x_1x_2 + 2x_1^2 + x_2^2$$

$$\nabla f(\mathbf{x}) = \begin{bmatrix} 1 + 2x_2 + 4x_1 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}$$

$$\mathbf{a} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

$$\nabla f(\mathbf{a}) = \begin{bmatrix} 15 \\ 11 \end{bmatrix}$$

$$\nabla^2 f(\mathbf{x}) = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$$

Summary

- Introduction to Number System
- Calculus of Single & Multiple Variables
- Partial Differentiation & Gradient
- Hessian Matrix



Thank You